

Aldehydes are organic compounds used as intermediates in metabolism and are also encountered in everyday life. Common aldehydes include formaldehyde, which is used in the fixation of tissues, and cinnamaldehyde, which provides the flavor and smell of cinnamon (Figure 1).

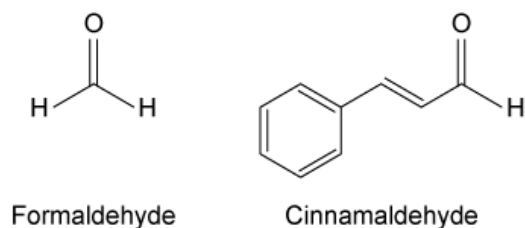


Figure 1 Common aldehydes

The aldol condensation shown in Figure 2 is an important carbon-carbon bond-forming reaction that has many applications in the synthesis of bioactive molecules. It also takes place in some metabolic processes, such as the citric acid cycle and gluconeogenesis. The aldol condensation begins with the aldol reaction, which can be acid- or base-catalyzed and requires nucleophilic addition to a carbonyl to provide the aldol product, which can be a β -hydroxy ketone or aldehyde. The final step of the aldol condensation consists of an acid- or base-catalyzed dehydration of the aldol product to yield an α,β -unsaturated ketone or aldehyde.

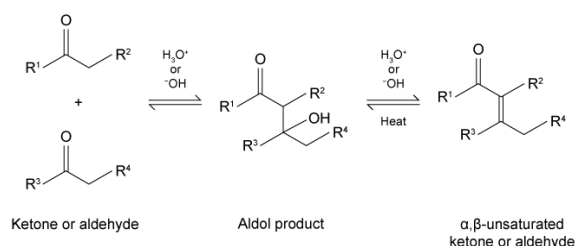


Figure 2 General aldol reaction and condensation

The reverse of an aldol reaction or condensation is known as a retro-aldol, which is important in metabolism, most notably in glycolysis. This reaction breaks a carbon-carbon bond to form aldehydes and/or ketones. In glycolysis, fructose 1,6-bisphosphate is broken down into an aldehyde and a ketone, glyceraldehyde 3-phosphate (GAP) and dihydroxyacetone phosphate (DHAP), respectively (Figure 3).

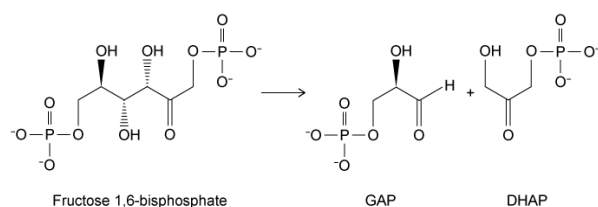
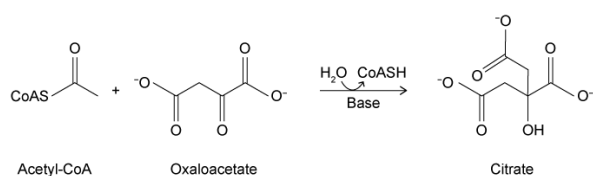
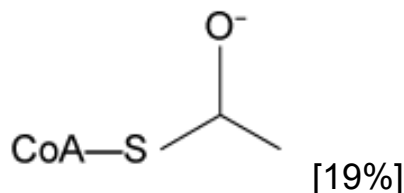


Figure 3 Retro-aldol in glycolysis

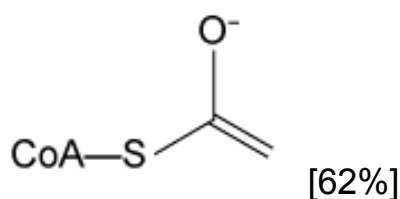
Which of the following compounds would be a plausible intermediate in the base-catalyzed aldol reaction of the first step of the citric acid cycle shown below?



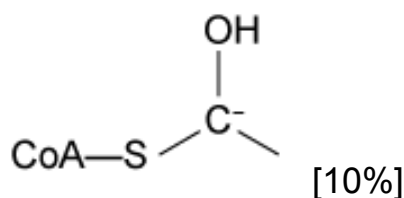
☐ A.



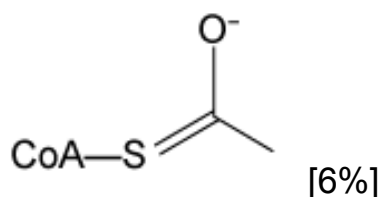
✓ ☒ B.



☐ C.



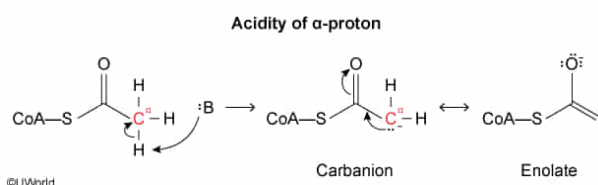
☐ D.



Correct

62% answered correctly

Explanation:



Protons on the carbon atom **adjacent (α) to a carbonyl ($\text{C}=\text{O}$)** group are more **acidic** than other protons bonded to a carbon atom for two reasons:

- The oxygen on the carbonyl group is electron-withdrawing, giving the carbonyl carbon a partial positive charge. The electrons of the $\text{C}-\text{H}$ bond are attracted toward the positive charge of the carbonyl carbon. The resulting lack of electron

density around the α -protons allows them to be **abstracted by** a strong **base**. When a proton is removed, the electrons from the C–H bond are transferred to the carbon atom, resulting in an anion.

- Charge delocalization over multiple atoms creates a more stable **resonance structure**. After deprotonation, the negative charge on the α -carbon can be delocalized to the oxygen atom of the carbonyl, forming an **enolate**. The enolate is more stable because oxygen is more **electronegative** than carbon and can more easily hold a charge.

In acetyl-CoA, the methyl ($-\text{CH}_3$) group is α to a type of carbonyl called a **thioester**, which makes it susceptible to deprotonation by a base. The resulting negative charge can be transferred from the α -carbon to the oxygen of the thioester, producing a stable enolate.

(Choice A) This structure does not show deprotonation of the methyl group of acetyl-CoA.

(Choice C) The negative charge is shown on the incorrect carbon atom. This structure would have to be an **enol** for the oxygen to be protonated.

(Choice D) This image shows a resonance structure that implies the sulfur from acetyl-CoA has been deprotonated; however, the sulfur atom does not have any protons to be abstracted. Sulfur also has only five electrons in this structure where it should have six valence electrons.

Educational objective:

Protons on the carbon atom adjacent (α) to a carbonyl are more acidic than other protons bonded to a carbon because they are electron-deficient and can be abstracted by a base to yield an anion. The charge is delocalized to form a more stable resonance structure known as an enolate.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401104

System:

Functional Groups and Their Reactions

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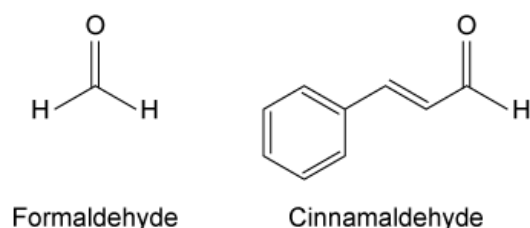


Figure 1 Common aldehydes

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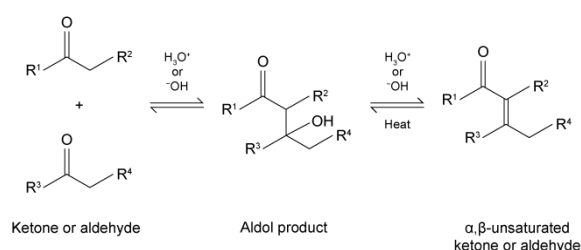


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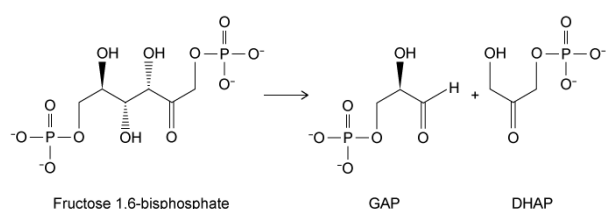


Figure 3 Retro-aldol in glycolysis

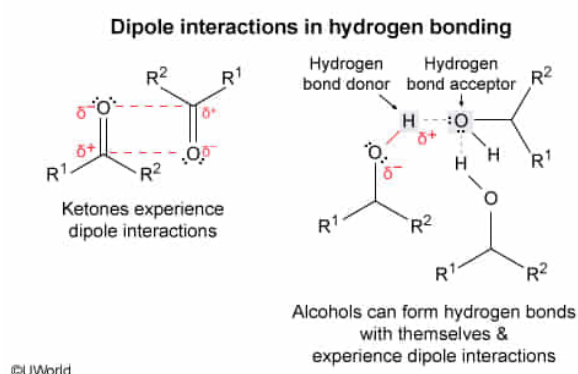
Do aldehydes and ketones tend to have higher or lower boiling points than their corresponding alcohols?

- ☐ A. Higher, because they are at a higher oxidation state [9%]
- ☐ B. Higher, because they have a dipole moment [8%]
- ☒ C. Lower, because they cannot form hydrogen bonds with each other [79%]
- ☐ D. Lower, because they have increased intermolecular forces [2%]

Correct

79% answered correctly

Explanation:



The boiling point of a substance depends on the strength of the **intermolecular forces** holding the molecules together. Stronger intermolecular forces require more energy to break, and therefore contribute to **higher boiling points**.

The electronegative oxygen in the hydroxyl ($-\text{OH}$) group of alcohols pulls electron density away from hydrogen, creating a **dipole** in which oxygen has a partial negative charge and hydrogen has a partial positive charge. Because opposite charges attract, the dipole provides an **increase in intermolecular forces**, and therefore an increase in boiling point relative to alkanes. In addition, the hydroxyl group can act as both a **hydrogen bond donor and hydrogen bond acceptor**, allowing alcohols to form **hydrogen bonds with themselves** and further increase the intermolecular forces.

Ketones and aldehydes contain carbonyls, in which the electronegative oxygen has a partial negative charge and the carbon has a partial positive charge. Like alcohols, ketones and aldehydes have dipoles and thus have stronger intermolecular interactions than alkanes. Unlike alcohols, aldehydes and ketones have hydrogen bond acceptors but not donors, so they **cannot form hydrogen bonds** with themselves. Therefore, they have **lower boiling points** than their corresponding alcohols.

(Choices A and B) An increased oxidation state and a dipole moment would cause the boiling point of ketones and aldehydes to increase with respect to their analogous alkanes but not sufficiently enough to exceed the boiling point of their corresponding alcohols, which can form hydrogen bonds.

(Choice D) Ketones and aldehydes have increased intermolecular forces relative to alkanes but not alcohols. Furthermore, the increased forces would contribute to raising the boiling point, not lowering it.

Educational objective:

Ketones and aldehydes contain carbonyls, which create dipoles and increased intermolecular forces. Because the oxygen of the carbonyl can accept but not donate hydrogen bonds, ketones and aldehydes cannot form hydrogen bonds with themselves. Alcohols have dipoles and can form hydrogen bonds with themselves, resulting in a higher boiling point than ketones and aldehydes.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401105

System:

Functional Groups and Their Reactions

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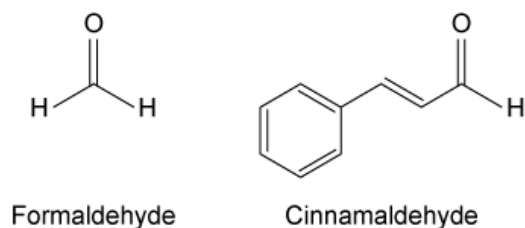


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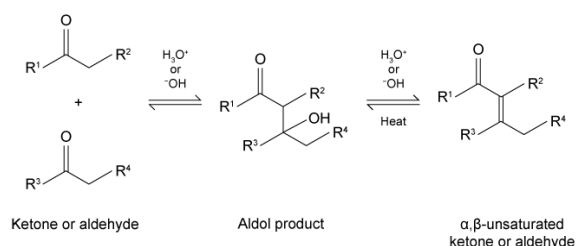


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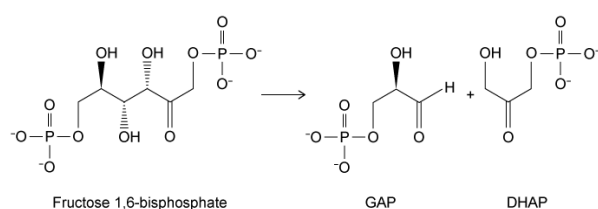
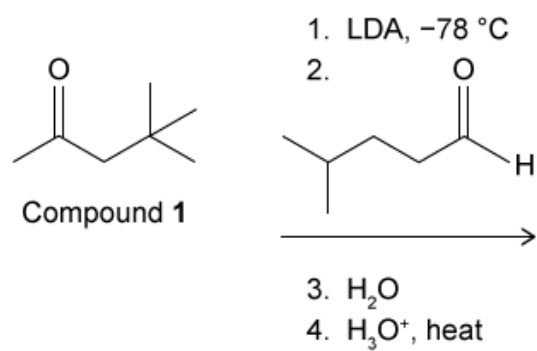


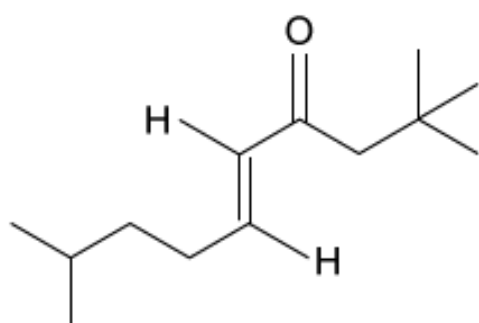
Figure 3 Retro-aldol in glycolysis

Consider the reaction detailed below.



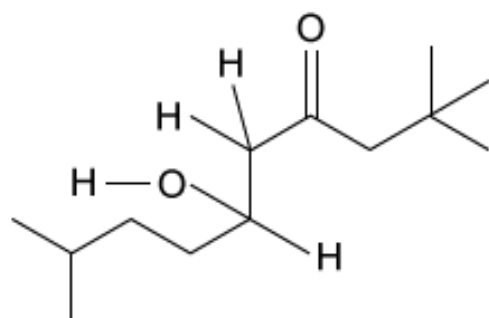
Which of the following would be the kinetic product if Compound **1** undergoes an aldol condensation as described in Figure 2?

✓ ☒ A.



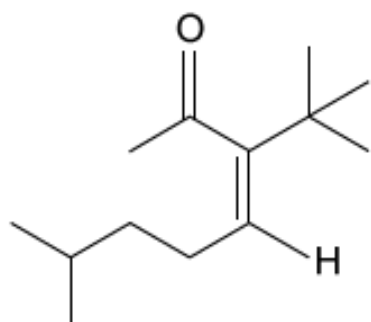
[46%]

☐ B.



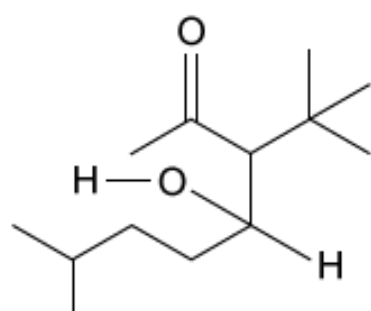
[12%]

☐ C.



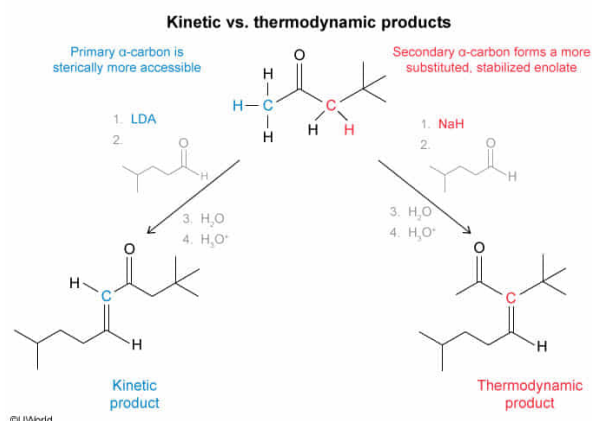
[27%]

☐ D.



Correct

45% answered correctly

Explanation:

The aldol condensation begins with enolate formation by deprotonation of a carbon atom adjacent to a carbonyl group ($\text{C}=\text{O}$), known as the α -carbon. A reaction at a **low temperature** (-78°C) with a **bulky base**, such as **lithium diisopropylamide (LDA)**, favors formation of the **kinetic** over the thermodynamic enolate because the kinetic enolate requires a low **activation energy** to form. The kinetic product is a result of deprotonation of the **less substituted α -carbon** (ie, contains more hydrogen substituents and is sterically *more accessible* to a bulky base). A reaction at a higher temperature with a smaller base such as sodium hydride (NaH) would lead to deprotonation of the *more* substituted α -carbon and formation of the thermodynamic enolate (ie, enolate $\text{C}=\text{C}$ double bond more stabilized by substituents) and therefore formation of the thermodynamic product.

This reaction begins with deprotonation of either the methylene ($-\text{CH}_2-$) group adjacent (α) to the carbonyl ($\text{C}=\text{O}$) and **tert-butyl group** or the methyl ($-\text{CH}_3$) α -carbon. Since the reaction is performed at a low temperature with the bulky base LDA, the less substituted α -carbon (the methyl group) is deprotonated **rapidly** and forms an anion. The negative charge is delocalized to the oxygen of the carbonyl, forming the **kinetic enolate**. The enolate acts as a **nucleophile** and **adds** to the carbonyl of the aldehyde, generating a negative charge on the carbonyl oxygen. Water **protonates** the negatively charged oxygen and forms the **aldol product**.

The final step is **acid-catalyzed dehydration**. Acid protonates the hydroxyl ($-\text{OH}$) group and gives oxygen a positive charge, making it a good **leaving group**. Water leaves and creates a **secondary carbocation**. Finally, water deprotonates the methylene group between the carbocation and the carbonyl, transferring the electrons from the $\text{C}-\text{H}$ bond to create a double bond and forming the **α,β -unsaturated product**.

(Choice B) This structure shows the *aldol product* after the kinetic enolate has formed but *not undergone dehydration*.

(Choices C and D) These structures show the dehydration and aldol products, respectively, after the thermodynamic enolate—but not the kinetic enolate—has formed.

Educational objective:

Kinetic enolates require low activation energy and form rapidly under low temperature and bulky base reaction conditions, and involve the deprotonation of the less-substituted α -carbon. The base-catalyzed aldol condensation has four steps: formation of an enolate, nucleophilic addition to a carbonyl, protonation, and acid- or base-catalyzed dehydration to form an α,β -unsaturated carbonyl.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401106

System:

Functional Groups and Their Reactions

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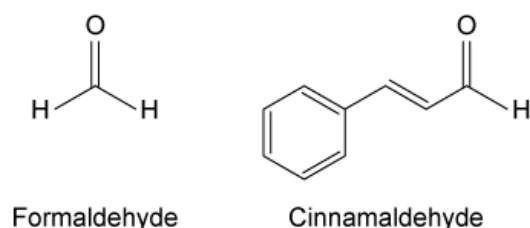


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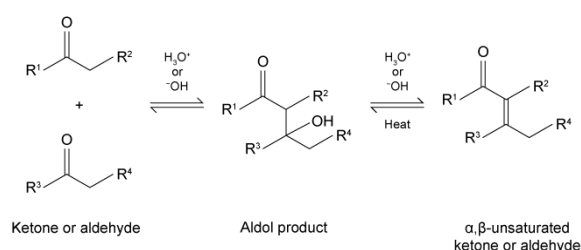


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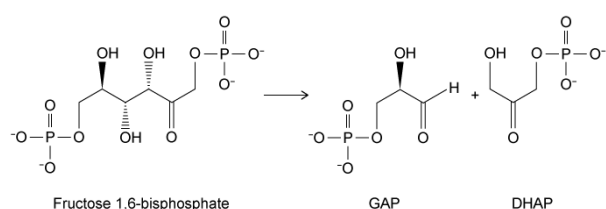
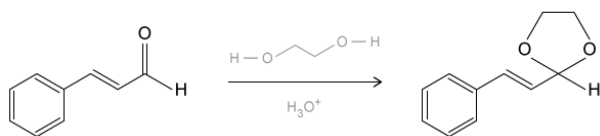


Figure 3 Retro-aldol in glycolysis

In some synthesis reactions, the carbonyl functional group needs to be protected from nucleophilic attack. The reaction below of cinnamaldehyde with ethylene glycol shows the formation of which carbonyl-protecting group?

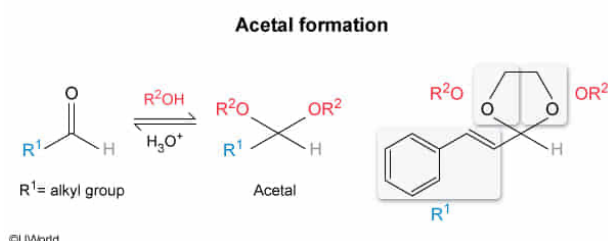


- ✓ ☒ A. Acetal [63%]
☐ B. Hemiacetal [24%]
☐ C. Ketal [6%]
☐ D. Hemiketal [6%]

Correct

62% answered correctly

Explanation:



Because ketones and aldehydes are more reactive than other carbonyl-containing **functional groups**, they sometimes need to be protected to prevent a reaction with nucleophiles. A common **protecting group** used for **aldehydes** is known as an **acetal**, which is characterized by **two –OR groups** in place of the carbonyl. This group can be distinguished from the ketal protecting group, which is formed by two –OR groups in place of a ketone rather than an aldehyde.

Nucleophilic addition of *one* equivalent of an alcohol to an aldehyde creates a **hemiacetal**, where **one –OR group** and **one –OH** are on the carbon atom that is part of the carbonyl. When *another equivalent* of alcohol is added, an acetal (ie, two –OR groups) is formed. The protecting group can later be removed via acid-catalyzed hydrolysis.

1,2- and 1,3-diols can also be used to form cyclic acetals. When cinnamaldehyde is treated with the diol ethylene glycol (HO(CH₂)₂OH) and acid, the carbonyl is protected as a cyclic acetal.

(Choices B and D) Hemiacetals and **hemiketals**, originating from aldehydes and ketones, respectively, each contain **one –OR group** and **one –OH group**.

(Choice C) Like an acetal, a **ketal** has two –OR groups in place of a carbonyl but contains *four* nonhydrogen substituents instead of three on the carbon atom that was originally the carbonyl group. It originates from a *ketone*, not an aldehyde.

Educational objective:

The carbonyl groups of aldehydes and ketones can be protected from nucleophiles by reaction with two equivalents of an alcohol to give an acetal and a ketal, respectively. Reaction with one alcohol equivalent (the halfway point) produces one –OR group and one –OH group in place of the carbonyl, generating a hemiacetal (for aldehydes) or hemiketal (for ketones).

Time spent:0 sec

Subject:

Organic Chemistry

QID:401107

System:

Functional Groups and Their Reactions

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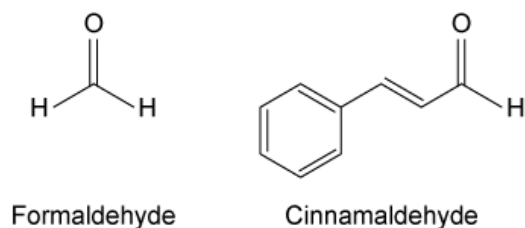


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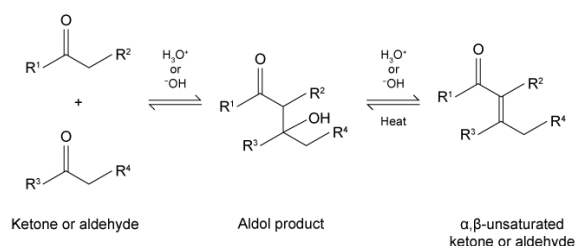


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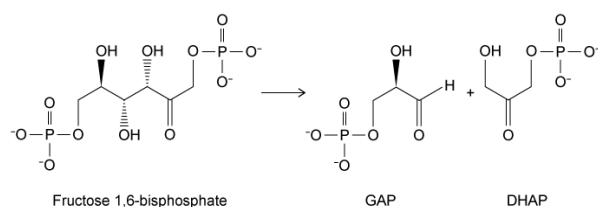


Figure 3 Retro-aldol in glycolysis

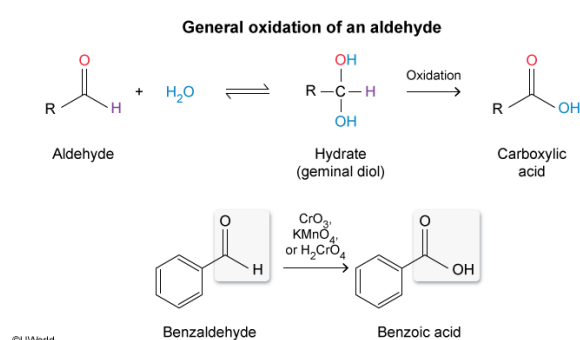
Which of the following reagents CANNOT be used to make benzoic acid from benzaldehyde?

- ☐ A. CrO_3 [6%]
☐ B. KMnO_4 [5%]
☐ C. H_2CrO_4 [7%]
✔ ☒ D. PCC [80%]

Correct

80% answered correctly

Explanation:



The **oxidation** of an aldehyde to a carboxylic acid requires an increase in the number of C–O bonds and a decrease in the number of C–H bonds. An increase in C–O bonds results in the loss of electron density around carbon and a partial positive charge while the oxidizing agent gains electrons and becomes reduced. **Aldehydes** are readily **oxidized to carboxylic acids** by numerous **oxidizing agents**, including **chromium reagents** and **potassium permanganate**, in an **aqueous environment**. The aldehyde is believed to go through a **hydrate intermediate**, also known as a geminal diol, before being oxidized to a carboxylic acid.

However, **pyridinium chlorochromate (PCC)** can only oxidize primary and secondary alcohols to aldehydes and ketones, respectively. PCC **cannot oxidize aldehydes further** because it is an **anhydrous** oxidizing reagent and does not contain the water necessary to convert the aldehyde to a hydrate, a necessary intermediate to be oxidized to the carboxylic acid.

(Choices A, B, and C) CrO_3 , KMnO_4 , and H_2CrO_4 can all oxidize aldehydes to carboxylic acids because these reagents are all in aqueous environments, meaning they contain the water necessary to convert the aldehyde to the hydrate intermediate that is oxidized to the carboxylic acid.

Educational objective:

Aldehydes are readily oxidized to carboxylic acids in aqueous environments by several oxidizing agents, including chromium compounds. Aldehydes are believed to go through a hydrate intermediate before being oxidized to a carboxylic acid. The anhydrous oxidizing reagent pyridinium chlorochromate (PCC) cannot oxidize aldehydes because it lacks the water needed to form the hydrate intermediate.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401108

System:

Functional Groups and Their Reactions

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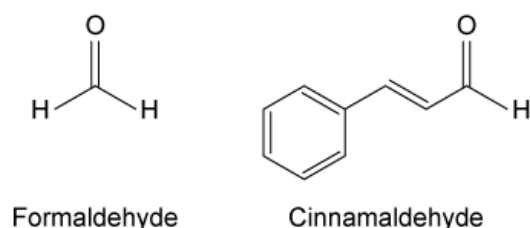


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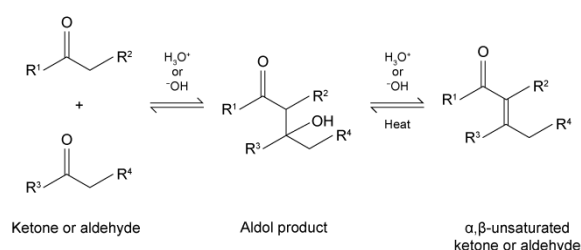


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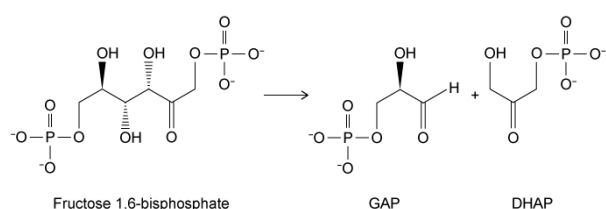
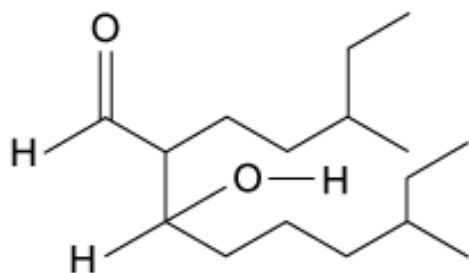


Figure 3 Retro-aldol in glycolysis

What is the product if the compound shown below undergoes a retro-aldol reaction?

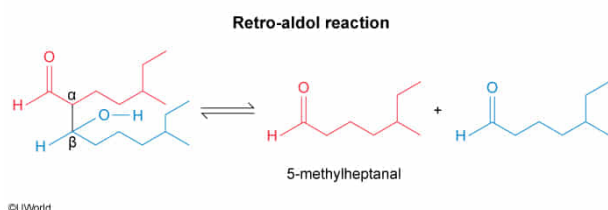


- ☐ A. 5-ethylhexanal [15%]
✔ ☒ B. 5-methylheptanal [67%]
☐ C. 5-ethylhexanone [7%]
☐ D. 5-methylheptanone [9%]

Correct

66% answered correctly

Explanation:



The **retro-aldol reaction** is the reverse of the aldol reaction or aldol condensation and is catalyzed by a base and heat. In these reactions, the **carbon-carbon bond** between the α - and β -carbon atoms is **broken**, yielding two aldehydes, two ketones, or one of each. In this case, two identical aldehydes are produced.

The following steps describe how to name an aldehyde:

1. Determine the **longest carbon chain that includes the aldehyde** functional group and begin numbering the chain at the carbonyl carbon. The longest chain in this reaction has seven carbon atoms.
2. Determine the name of each substituent, if any. All substituent names are placed at the beginning of the compound name together with the number of the carbon to which the substituent is attached. In this case, the methyl group at carbon 5 is listed as "5-methyl."

3. Name the **parent alkane** (longest chain) and replace the –e at the end with the **suffix –al** to indicate the aldehyde functional group. In this case, the name of the compound is **5-methylheptanal**.

(Choices A and C) For the names 5-ethylhexanal and 5-ethylhexanone, the longest carbon chain was determined incorrectly. Therefore, the substituent (ethyl) and parent name prefix (hex) are incorrect.

(Choices C and D) For the names 5-ethylhexanone and 5-methylheptanone, the highest priority functional group was incorrectly named. The suffix **–one** refers to a ketone, not an aldehyde.

Educational objective:

The retro-aldol reaction is the reverse of the aldol reaction, in which the bond between the α - and β -carbon is broken into two fragments. Aldehydes are named after the number of carbon atoms in the longest carbon chain, replacing the –e in the parent alkane name with the suffix **–al**. Substituent names precede the chain name and are numbered from the carbonyl carbon.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401109

System:

Functional Groups and Their Reactions